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PHYSICS

Ans 1 When a light bulb is placed on the focus of a concave mirror the parallel beam will be its reflected beam.

Ans 2 $R = 2f$
 $\Rightarrow f = \frac{R}{2}$

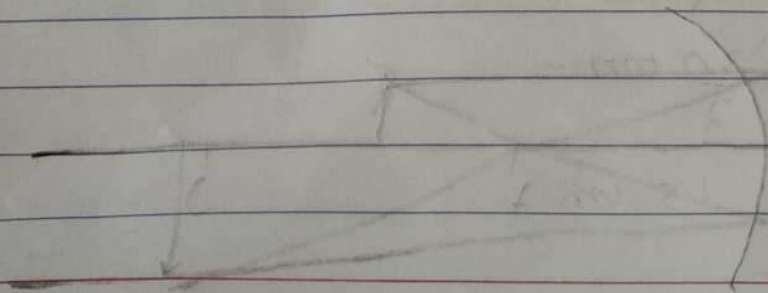
$\Rightarrow R = 50 \text{ cm}$

$\Rightarrow f = \frac{50}{2}$

$\Rightarrow f = 25 \text{ cm}$

It will provide convex mirror.

Ans 3 We can see the image for the formation of concave mirror. When object is placed between pole C and F.



Ques 4 Given

$$u = -15 \text{ cm}$$

$$f = 10 \text{ cm}$$

$$\text{height of object} = 3 \text{ cm}$$

$$v = ?$$

$$h_1 = ?$$

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

$$\Rightarrow \frac{1}{10} = \frac{1}{v} + \left(\frac{1}{-15}\right)$$

$$\Rightarrow \frac{1}{10} + \frac{1}{15} = \frac{1}{v}$$

$$\Rightarrow \frac{3 + 2}{30} = \frac{1}{v}$$

$$\Rightarrow \frac{5}{30} = \frac{1}{v}$$

$$\Rightarrow v = \frac{30 \text{ cm}}{5}$$

$$\Rightarrow v = 6 \text{ cm}$$

$$m = \frac{h_2}{h_1} = -\frac{4}{4}$$

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$$\Rightarrow \frac{h_2}{3} = \frac{+(-30)}{+15} = \frac{+(-15)}{+(-15)}$$

$$\Rightarrow h_2 = 1 \times 3$$

$$\Rightarrow h_2 = 4 \text{ cm} \quad 3 \text{ cm}$$